103

Multiligament Knee Injuries

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Knee ligaments are responsible for providing the static stability of the knee, control of kinematics, and prevention of abnormal rotation and/or displacement that may damage the articular surfaces or the menisci. Knee dislocations are rare and are estimated to account for 0.02% to 0.2% of orthopaedic injuries[1]; however, it is generally accepted that multiligament injuries may occur at a higher rate because some knees spontaneously reduce before presentation.[2,3] These injuries may spontaneously reduce or may present as an acutely dislocated knee requiring reduction. The diagnosis and management of multiligament injuries pose unique challenges to orthopaedic surgeons, and a wide spectrum of injury exists, ranging from two-ligament injuries such as a cruciate and collateral ligament rupture to a grossly unstable knee that requires spanning external fixation. Although the immediate concerns should be to determine the integrity of the neurovascular structures, other essential concepts include accurate identification of all injured structures, repair versus reconstruction, management of acute versus chronic injuries, single- versus two-stage surgery, and postoperative rehabilitation. During the past three decades, clinical outcome studies along with anatomic and biomechanical investigations have greatly improved the management of these complex injuries.

HISTORY

In evaluating a patient who presents with knee pain or instability, the clinician must obtain a careful history of symptom onset, mechanism of injury, history of prior knee injuries, and previous operative and nonoperative treatments. Multiligament injuries associated with sports are considered low-energy and are often isolated to the involved extremity, whereas those associated with automobile or motorcycle crashes are considered high-energy[4] and may be combined with other life-threatening injuries.

Acutely injured patients may be unable to ambulate because of swelling, pain, and instability. Determination of the time since the injury occurred is crucial in patients who present with persistent dislocation because of the possibility of vascular injury and limb ischemia. Patients with chronic injuries may report mechanical symptoms, including clicking, catching, or locking or they may report instability on uneven ground, with cutting motions, and during activities of daily living. Neurologic deficits may be reported, including the presence of paresthesias in the common peroneal nerve distribution and a foot drop. Synthesis of this information will guide the clinician in the physical examination and selection of imaging studies.

PHYSICAL EXAMINATION

Examination of a patient with a suspected multiligament injury in the acute setting must include the assessment of vascular status. If an arterial injury is suspected, an ankle-brachial index score should be determined; a score of less than 0.9 is an indication that advanced arterial imaging should be obtained.[5] Serial neurovascular examination and selective computed tomography angiography has been recommended; “hard signs” of ischemia warrant emergent vascular consultation.[6]

The neurologic status must also be assessed. The common peroneal nerve supplies motor innervation to the anterior (deep peroneal) and lateral (superficial peroneal) compartments of the leg, as well as the extensor hallucis brevis and extensor digitorum brevis (deep peroneal) on the dorsum of the foot. A 25% to 35% nerve injury rate has been reported in the population with high-velocity knee dislocations.[7] In a series of acute isolated or combined posterolateral corner (PLC) knee injuries in an orthopaedic sports medicine referral practice, 4 of 29 patients had a complete palsy of the common peroneal nerve and an additional 7 of 29 had a partial motor/sensory deficit.[8] A recent study by Moatshe et al. found that peroneal nerve injury was significantly associated with vascular injury; thus injury to the peroneal nerve should raise suspicion of a vascular lesion.[2] Tibial nerve injuries may also occur in knee dislocations; although these injuries occur less frequently.

Physical examination of knee stability is a repeatable method of predicting intra-articular pathology but may be more difficult in patients with acute injuries. It is important to examine both legs to assess for pathologic instability versus physiologic laxity. Multiligament injuries are not subtle on examination; however, attention to subtle findings will aid the clinician in determining which specific structures are injured. Anteroposterior (AP) stability should be assessed with the Lachman, pivot shift, and posterior drawer tests. The posterior sag and quadriceps active tests also aid in the evaluation of the posterior cruciate ligament (PCL).

Lateral and posterolateral knee injuries are typically combined with an injury to one or both of the cruciate ligaments.[8-10] In acute injuries, the patient may have tenderness upon palpation of the fibular head. Examination maneuvers should include varus stress[11] at 0 and 20 degrees, reverse pivot shift, external rotation

Abstract

Knee ligaments are responsible for providing the static stability of the knee, control of kinematics, and prevention of abnormal rotation and/or displacement that may damage the articular surfaces or the menisci. Knee dislocations are rare and are estimated to account for 0.02% to 0.2% of orthopaedic injuries; however, it is generally accepted that multiligament injuries may occur at a higher rate because some knees spontaneously reduce before presentation. These injuries may spontaneously reduce or may present as an acutely dislocated knee requiring reduction. The diagnosis and management of multiligament injuries pose unique challenges to orthopaedic surgeons, and a wide spectrum of injury exists, ranging from two-ligament injuries such as a cruciate and collateral ligament rupture to a grossly unstable knee that requires spanning external fixation. Although the immediate concerns should be to determine the integrity of the neurovascular structures, other essential concepts include accurate identification of all injured structures, repair versus reconstruction, management of acute versus chronic injuries, single- versus two-stage surgery, and postoperative rehabilitation. During the past three decades, clinical outcome studies along with anatomic and biomechanical investigations have greatly improved the management of these complex injuries.

Keywords

multiligament
knee dislocation
knee injury
knee reconstruction
PLC
PCL

recurvatum,[12] and the dial test at 30 and 90 degrees. In a patient with a positive varus stress test at 30 degrees and negative findings at 0 degrees, an isolated fibular collateral ligament (FCL) tear is suspected. However, with multiligament injuries, the varus stress test will also be positive at 0 degrees. A positive dial test at both 30 and 90 degrees suggests a combined PCL and PLC injury.[13,14] A positive posterolateral drawer test reinforces findings consistent with a PLC injury but must be interpreted with caution, as discussed further on.

Medial structures are evaluated with the valgus stress test at 0 and 20 degrees of flexion[15]; instability at full extension is indicative of a combined cruciate ligament injury. Assessment of medial compartment gapping at 20 degrees under a valgus stress primarily isolates the superficial medial collateral ligament (MCL). Evaluation of associated rotational abnormalities is assessed with anteromedial tibial rotation at 90 degrees of flexion and the dial test at 30 and 90 degrees of flexion.[16] Increased anteromedial rotation suggests a more extensive knee injury that includes the superficial MCL as well as the posterior oblique ligament (POL) and deep MCL. The examiner must be careful to differentiate anteromedial from posterolateral tibial rotation during the dial test by palpation and visualization of tibial subluxation with the patient in the supine position.[16]

Gait assessment is an important component of the physical examination but may be compromised because of pain in persons with acute injuries. In subacute or chronic injuries, a varus thrust gait or foot drop may be observed in patients with combined lateral injuries. Patients with medial knee injuries may demonstrate a valgus thrust during the stance phase of gait, but this manifestation is less common and usually occurs in patients with genu valgus alignment.

IMAGING

Radiographic examination for patients with a suspected multiligament knee injury should include standard AP and lateral views (Fig. 103.1) as well as weight-bearing flexion (Rosenberg) views.[17] These views allow visualization of tibial plateau, femoral condyle, or osteochondral fractures. Segond[18] and/or arcuate[19] fractures may be visualized with lateral/posterolateral injuries, and calcification near the MCL origin (Pellegrini-Stieda ossification) may be visualized in chronic medial-sided injuries. Baseline bilateral standing long-leg radiographs allow the clinician to determine the mechanical axis of the injured and contralateral extremities, which may have a significant impact on treatment decisions for chronic multiligament injuries.[20,21]

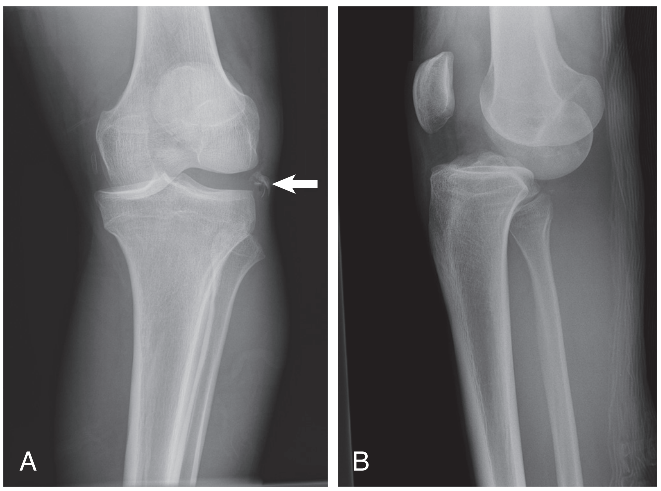

Fig. 103.1 Anteroposterior (A) and lateral (B) radiographs demonstrating an acute left knee dislocation. An arcuate fracture is also visible on the anteroposterior radiograph *(arrow).*

Preoperative stress radiographs provide quantitative objective information on the stability to valgus and varus stress and should also be routinely obtained postoperatively. Biomechanical studies were performed to objectively quantify the amount of joint opening with varus[11] and valgus[15] stress; radiographic techniques were developed and tested by sequential sectioning in cadaveric knees with intact cruciate ligaments. Isolated sectioning of the FCL (simulating a grade III injury) resulted in an increase of 2.7 mm of lateral joint gapping at 20 degrees of flexion when compared with the contralateral knee. Sectioning of the FCL, popliteus tendon (PLT), and PFL (simulating a complete grade III PLC injury) was associated with lateral joint gapping of 4 mm at 20 degrees of flexion. Isolated sectioning of the superficial MCL (simulating a grade III injury) resulted in 3.2 mm of increased medial joint gapping at 20 degrees of flexion when compared with the contralateral knee. Increased medial joint gapping of 6.5 and 9.8 mm at 0 and 20 degrees of flexion, respectively, was associated with sectioning of the superficial MCL, deep MCL, and POL (simulating a complete medial knee injury).

Several imaging techniques have been developed to allow quantitative assessment of the integrity of the PCL; these are especially useful in persons with chronic injuries. Stress radiographs have been described using the kneeling knee technique[22] and Telos device (Telos GmbH, Marburg, Germany)[23]; these two techniques have been reported to allow the quantification of posterior displacement of the tibia and are superior to a physical examination and use of the KT-1000 arthrometer.[24]

Magnetic resonance imaging (MRI) has become part of the standard of care for the evaluation of knee instability, especially in persons with acute injuries for whom examination may be limited by pain and swelling (Fig. 103.2). With high sensitivity and accuracy, MRI allows visualization of the cruciate and collateral ligaments, posteromedial corner and PLC,[25] bone marrow edema,[9,26,27] meniscal injuries, and cartilage lesions. Anteromedial femoral condyle bone bruises should alert the physician to a possible PLC injury.[9] In addition, lateral-sided tibial or femoral bone bruises have also been associated with MCL injuries.[28]

It is important to recognize common imaging findings associated with multiligament knee injuries. Plain radiographs may be negative in the acute setting if the patient is lying supine, and it may be difficult to obtain weight-bearing films. In chronic injuries, weight-bearing films may reveal varus or valgus gapping or loss of joint space and early findings of degenerative disease. Stress radiographs are especially useful to quantitatively assess stability of the PCL, medial complex, and posterolateral complex. In patients with acute injuries, MRI will usually reveal the status of the cruciate ligaments and allow assessment of the collateral

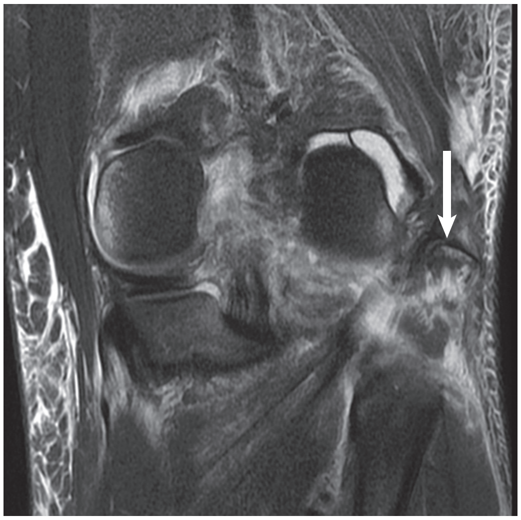

Fig. 103.2 A coronal magnetic resonance image of a left knee injury with bicruciate rupture and a complete posterolateral corner injury including arcuate fracture *(arrow)* and grade III injury of the fibular collateral ligament, popliteus tendon, and popliteofibular ligament.

structures if appropriate cuts and slice thicknesses are obtained. Classic bone bruises associated with ACL ruptures may be seen, as well as those associated with PLC injuries.

DECISION-MAKING PRINCIPLES

Patients with multiligament injuries are a heterogeneous group and may present with a variety of skin, bony, neurovascular, and ligamentous injuries. Although several general treatment algorithms have been developed, individualized treatment for the patient's specific knee injuries and concomitant injuries is necessary. Meniscus injuries should ideally be repaired, especially in the young patient, but this may require a partial meniscectomy. Management of vascular injuries, open injuries, skin coverage, fracture treatment, and meniscus injuries is not specifically discussed here. Important considerations for the treatment of multiligament injuries include operative versus nonoperative treatment, surgical timing, single- versus two-stage cruciate ligament reconstruction techniques, and repair versus reconstruction of collateral structures.

Knee ligament injuries have historically been classified with the use of a grading scale that assesses sagittal (AP) and coronal (varus/valgus) plane[16,29] stability. Rotational stability does not have a formal classification system, although many injury types have been described.[30-32] Treatment must be based on the extent of injury to individual structures and the number of structures injured. Knee ligament injuries are often subjectively classified according to the original American Medical Association guidelines, rated as grade I, II, or III.[33] An additional classification system is based on the number and location of torn ligaments.[34]

Nonoperative Versus Operative Treatment

It must be recognized that multiligament knee injuries are rare and that few studies have been conducted that compare treatment strategies with a high level of evidence. Current literature favors surgical management of multiligament knee injuries,[35-37] whereas in early reports nonoperative treatment was often recommended for "uncomplicated" cases (i.e., absence of vascular injury or fracture).[38,39] A recent review indicates improved outcomes in multiple subjective and objective facets for patients treated with an operation compared with those treated conservatively with immobilization.[37]

Surgical Timing

Several studies have evaluated the impact of surgical timing. However, interpretation of outcomes of surgically treated multiligament injuries is difficult because of the wide range of pathology within individual studies.[36] Irreducible knee injuries, open injuries, and popliteal vascular injuries necessitate emergent management. If the multiligament injury is associated with a high-energy trauma, the patient's overall medical status and serious concomitant extremity, torso, and head injuries may delay definitive treatment. Overlying skin injuries and associated plateau or femoral condyle fractures may necessitate delayed ligament reconstruction. These complicating factors are not specifically evaluated; rather, the focus here is on single-extremity multiligament injuries without concomitant injuries.

Timing of surgery is typically divided into one of the following three categories: acute (often defined as surgery within 6 weeks), chronic (often defined as surgery after 6 weeks), or staged (the index procedure is performed within 3 to 6 weeks of injury and second-stage surgery is delayed).[36] Harner et al.[40] reported improved subjective outcomes in acutely treated patients and no ultimate difference in range of motion (ROM); however, 4 of 19 patients with acutely reconstructed knees required manipulation for loss of flexion. Fanelli and Edson[41] reported on 35 patients with multiligament injuries and found no subjective differences (according to Tegner, Lysholm, and Hospital for Special Surgery knee ligament rating scales) or objective differences (according to use of the KT-1000 arthrometer) between the acute and chronic cohorts.

A recent systematic review of surgical treatment for multiligament injuries found increased anterior instability for patients treated acutely but no difference in posterior, varus, or valgus instability when compared with chronically treated injuries.[36] Additionally, flexion loss (>10 degrees) as well as the need to undergo a subsequent repeat operation for stiffness were more frequent in acutely treated patients; no difference was found for extension. Last, patients treated with staged reconstructions had more "excellent" or "good" outcomes than did those treated acutely. As described later in this chapter, these findings may be difficult to interpret because many of these patients were treated with acute PLC repairs (which have been found to frequently fail) rather than reconstructions.

Patients with chronic multiligament injuries may present to the orthopaedic surgeon because of a failed index procedure, failed nonoperative management, or concomitant injuries that precluded acute surgical management of the multiligament injury. These patients may have varus malalignment, and a high-tibial osteotomy may be required to correct the mechanical axis prior to ligament reconstruction because a failure to correct varus malalignment with a chronic PLC injury has been reported as a cause of PLC reconstruction graft failure.[42,43]

Cruciate Ligament Reconstruction

It is widely accepted that cruciate ligament injuries in patients with multiligament injuries require reconstruction. A biomechanical study by Veltri and colleagues[44] demonstrated the importance of reconstructing the cruciate ligaments in multiligament injuries. Options for anterior cruciate ligament (ACL) reconstruction in multiligament injuries include transtibial versus transportal drilling for femoral tunnels and autograft versus allograft; no known studies recommend double-bundle ACL reconstruction in these patients. The debate regarding the best PCL reconstruction technique to use in persons with multiligament injuries is similar to the debate for isolated injuries; graft fixation techniques, single- versus double-bundle technique, and transtibial tunnel versus tibial inlay technique. Few studies have been performed to compare cruciate ligament reconstruction techniques in the multiligament injury patient population; therefore surgeons must apply the principles used for the reconstruction of isolated cruciate ligament injuries to this unique patient group.

Collateral Structures

Until recently it was believed that PLC structures could be successfully repaired if treated acutely. This practice has been challenged by outcomes studies that compared repairs versus reconstructions.[45,46] It has been biomechanically demonstrated that a deficient PLC leads to increased ACL[47] and PCL[48,49] graft forces; interestingly, Mook et al.[36] reported that more patients treated acutely for multiligament injuries underwent repairs rather than reconstructions of the PLC and suggest that repairs may have been insufficient to protect the ACL graft during healing. These findings may provide clinical evidence that reinforces the biomechanical principles of secondary stabilization between cruciate and collateral ligaments and may support the trend toward acute reconstruction rather than repair of PLC structures.[50] As discussed later in this chapter, a gradual trend has occurred from local tissue transfers and acute repairs toward several different autograft or allograft tissue reconstruction techniques.

A well-defined and successful treatment algorithm for isolated grade III MCL injuries and those combined with ACL ruptures includes a short period of rest and edema control followed by physical therapy.[16] However, treatment of MCL injuries associated with bicruciate injuries is less well defined. Some authors advocate delayed cruciate reconstruction while the medial structures are protected with a brace and allowed to potentially heal. Other authors recommend acute repairs or reconstruction of medial structures, although a higher risk of arthrofibrosis is reported.

Graft Choice

Graft choice in multiligament injury reconstruction is determined by injury pattern, graft availability, and surgeon preference. Often surgeons prefer to use allografts when treating multiligament injuries because of multiple graft size options and the ability to avoid the increased operative time and donor site morbidity associated with harvesting the patellar and hamstring tendon and possibly quadriceps tendon grafts. Because of the heterogeneity of multiligament injuries, no conclusive studies are available to recommend a particular graft choice. Several techniques are discussed in the Treatment Options section, further on, along with graft choices.

Special Considerations

Pediatric patients with multiligament injuries require special consideration because of their open physes and the potential risk of growth alteration with traditional ligament reconstruction. Physeal sparing techniques for ACL[51] and PCL[52] reconstruction have been described. Recently a modified physeal sparing technique was described for PLC/FCL injuries in pediatric patients.[53] Lateral and medial structures may be repaired via augmentation or recess procedures[8] or with use of suture anchors.

A recently defined type of knee dislocation described by Azar and colleagues has been termed *ultra–low velocity*.[54] These injuries are sustained by patients with a high body mass index as a result, for example, of falling from a standing height or tripping on an object. Patients who sustain ultra–low velocity multiligament knee injuries have higher rates of neurovascular injuries as well as a higher incidence of postoperative complications compared with patients who suffer low- or high-velocity multiligament injuries.[55] Their treatment must be individualized based on medical comorbidities, patient expectations, preinjury activity level, and ability to comply with rigorous rehabilitation.

Conservative therapy with immobilization may be the only treatment suitable for elderly patients with multiligament injuries who have preinjury medical comorbidities.[56] Additionally, the presence of arthritis is a relative contraindication to multiligament reconstruction; in fact, most studies exclude patients with preexisting arthritis.

TREATMENT OPTIONS

Treatment recommendations for specific ligament injuries in this patient population are limited by the lack of comparative studies. No known studies have evaluated the impact of a specific cruciate ligament reconstruction technique on the outcomes of multiligament injuries. Repair versus reconstruction of collateral ligaments has been debated, but specific reconstruction methods have not been directly compared in clinical studies.

Acute Management

Every multiligament knee injury is unique, and a wide range of pathology exists. Most injuries are adequately stabilized in a knee immobilizer. However, some knees associated with a high-energy injury may remain subluxed in a knee immobilizer and require a spanning external fixator to achieve stability in the acute setting.

Anterior Cruciate Ligament

Although ACL reconstruction is recommended, the specific technique receives relatively little discussion in the context of multiligament injuries. Many authors have described single-bundle reconstruction using an allograft or autograft with femoral tunnels created via a transtibial technique. Levy et al.[45] prefer to use a tibialis anterior allograft, Strobel et al.[57] recommended use of a hamstring autograft. Engebretsen et al.[1] initially preferred allografts for ACL reconstruction but changed their graft choice to a bone–patellar tendon–bone autograft. Harner et al.[40] prefer allograft bone–patellar tendon–bone but use an anteromedial portal drilling technique for ACL femoral tunnels rather than the transtibial technique as utilized by previous authors.

Posterior Cruciate Ligament

A review of treatment options for addressing PCL insufficiency in this patient population follows. It is generally accepted that PCL tears should be reconstructed in these patients, although the optimal technique has not yet been defined. A review of the causes of failure of a series of PCL reconstructions identified the importance of tunnel positioning and addressing concomitant collateral ligament instability.[21] However, investigators have not yet determined the role for the single- versus double-bundle technique and for transtibial versus tibial inlay graft placement.

Some studies have demonstrated that double-bundle PCL reconstructions restore native biomechanics[48,58]; however, relatively few studies have specifically described the detailed technique and associated outcomes of double-bundle PCL reconstructions in the multiligament injury population. Spiridonov et al.[59] recently described a double-bundle PCL reconstruction in 7 patients with isolated PCL ruptures and 32 with multiligament injuries. Their technique includes two femoral tunnels and a single transtibial tunnel to anatomically reconstruct the anterolateral bundle (ALB) and posteromedial bundle (PMB). Because of the morbidity of graft harvest and the need for large collagen volume, the authors used allografts, specifically Achilles tendon, for the ALB and a semitendinosus tendon for the PMB.

Several authors have described single-bundle PCL reconstructions in patients with multiligament injuries. Fanelli and Edson[41] recommended a single-bundle transtibial PCL reconstruction and reported using either an autograft or allograft. Engebretsen et al.[1] described a single-bundle transtibial PCL reconstruction; during the time of data collection, the investigators changed their graft source from allograft to hamstring autograft. Chhabra et al.[60] reported that approximately one-third of patients with an acute multiligament injury have an intact PMB and meniscofemoral ligaments. They attempted to preserve these bundles and performed a reconstruction of the ALB using an Achilles tendon allograft via a transtibial tunnel. In patients with complete ruptures of the entire PCL and those with chronic injuries, the authors recommend double-bundle PCL reconstruction using an Achilles tendon allograft for the ALB and a tibialis anterior allograft for the PMB.

A recent systematic review on the topic of transtibial versus tibial inlay technique for PCL reconstruction revealed a paucity of comparative outcomes studies and recommended surgeon preference as a reasonable consideration in technique choice until further evidence is available.[61] Biomechanical studies have compared the two techniques but it is difficult to apply their results to the multiligament injury population. Some investigators have recommended that the tibial inlay technique not be used for patients with multiligament injuries; however, this evidence is level V.[62] Stannard et al.[63] described a technique for double-bundle PCL reconstruction with a tibial inlay technique in patients with multiligament injuries. Their technique requires a single Achilles tendon allograft, split longitudinally, to reconstruct the ALB and PMB. Cooper and Stewart[64] described a single-bundle tibial inlay PCL reconstruction using either a bone–patellar tendon–bone autograft or allograft to reconstruct the ALB.

Medial/Posteromedial Structures

In a recent review, LaPrade and colleagues[16,65] underscored the importance of completely evaluating and treating the three main structures of the medial/posteromedial knee: the superficial MCL, deep MCL, and POL. When associated with multiligament injuries, most authors agree that grade III MCL injuries require treatment, often repair or reconstruction. A systematic review reported an absence of sufficient studies to allow formulation of evidence-based recommendations for the treatment of MCL injuries in the multiligament injury population.[66] A general trend has been noted in the literature toward repair and/or reconstruction of medial knee injuries, which may be due to a better understanding of the anatomy and availability of biomechanically validated reconstruction techniques.

In early reports, Fanelli et al.[67] compared valgus stability in patients with bicruciate ruptures and medial-sided injuries. Bicruciate reconstructions were performed in all patients; two acutely presenting patients were treated with primary surgical repair of the MCL tears and seven were treated with bracing to allow the MCL injury to heal with nonoperative treatment followed by subsequent bicruciate ligament reconstruction. More recently, Fanelli and Edson[41] describe an anterosuperior shift of the posteromedial capsule for the repair of MCL injuries; when lesions are not amenable for repair, an autograft semitendinosus or allograft is used to reconstruct the superficial MCL and is accompanied by a capsular shift.

Lind et al.[68] described a rerouting of the ipsilateral semitendinosus tendon to reconstruct the superficial MCL and posteromedial structures in patients with multiligament injuries. The semitendinosus tendon was identified and harvested proximally but left intact at the pes insertion. A blind femoral tunnel, located at the isometric point of the MCL insertion, was created with a diameter equal to the size of the double-looped tendon. Additionally, a transtibial tunnel exiting 10 mm below the tibial plateau and posterolateral to the semimembranosus was drilled through the medial tibial plateau from anterior to posterior and reamed to the diameter of the semitendinosus tendon. The double-looped tendon was secured using interference screw fixation in the femur, pulled through the tibial tunnel, and secured with an additional interference screw.

LaPrade and colleagues[65] described an anatomically based and biomechanically validated[69] reconstruction of the medial knee structures.[70] Their technique reconstructs the POL and both the proximal and distal divisions of the superficial MCL. Two femoral and two tibial tunnels are created, and grafts are fixed in the tunnels with use of interference screws.

In a series of patients with knee dislocations, Harner et al.[40] described repair or reconstruction of MCL injuries. Avulsions and midsubstance injuries were repaired with suture anchors and nonabsorbable sutures, respectively. Chronic injuries were treated with a reconstruction of the MCL using a semitendinosus autograft or Achilles tendon allograft.

Lateral/Posterolateral Structures

In contrast to the extra-articular medial structures, it is well recognized that grade III PLC injuries do not heal with bracing and, without operative treatment, can lead to significant morbidity. Recently reconstruction rather than repair of PLC injuries has been emphasized because of results of comparative outcomes studies. Early investigators reported good results with acute anatomic repair of PLC injuries; however, these patients were immobilized in a cast for 6 weeks postoperatively, and subjective outcomes scoring tools were not available.[71-73]

More recently, Stannard et al.[46] and Levy et al.[45] performed a mix of single- and dual-stage operations and found lower failure rates with reconstructions when compared with repairs of the PLC. Stannard et al.[46] performed a modified two-tailed technique with a tibialis anterior or posterior allograft for PLC reconstructions. This technique uses a single femoral tunnel at the isometric point along with a single fibular and tibial tunnel. Levy et al.[45] used a fibula-based technique with an Achilles tendon allograft, along with an anterodistal shift of the posterolateral capsule, to reconstruct the PLC.

LaPrade and colleagues recently reported on the acute[8] and chronic[20] treatment of isolated and combined PLC injuries. All PLC and concomitant cruciate ligament tears were treated with a single-stage surgery. Acute avulsions of PLC structures were repaired with suture anchors or recess procedures; however, most acute PLC injuries were not amenable to repair and were treated with a complete anatomic PLC reconstruction of the FCL, PLT, and/or PFL. A minority of the patients in the chronic PLC injury study were found to have varus malalignment and required an opening wedge proximal tibial osteotomy to correct the mechanical axis before undergoing a soft tissue reconstruction. The remainder of the patients were treated with an anatomic reconstruction of the PLC with single-stage reconstruction of coexistent cruciate ligament tears.

Authors' Preferred Technique

Ligamentous Injuries

Our preferred technique for the treatment of multiligament injuries is an anatomic single-stage reconstruction of the cruciate ligament(s) with concurrent treatment of medial/posteromedial and lateral/posterolateral supporting structures with anatomically based and biomechanically validated techniques. Grade III injuries to the medial and lateral structures require surgical treatment for patients with multiligament injuries with a repair and/or reconstruction when indicated. A repair of some structures may be possible in acute injuries with avulsions directly off bone; however, a reconstruction is required for acute injuries with midsubstance tears or inadequate tissue quality and for chronic injuries. It is the preference of the senior author to operate on patients with acute injuries within 3 weeks of injury so as to allow identification of injured structures and repair of meniscal pathology and extra-articular structures.

Preoperative Planning

Preoperative planning for the treatment of multiligament injuries is critical because of the inherent complexity of these procedures. The injury history, physical examination, and imaging studies will allow the surgeon to plan the details of the operation. The surgeon must be certain that all required equipment is available, including surgical instruments and any required allograft materials. Standard cruciate ligament reconstruction instruments including cannulated drill guides, eyelet-tipped passing pins, suture anchors, and cannulated interference screws (metallic or bioabsorbable) will be necessary. Appropriate graft harvesting instruments will be needed if the surgeon plans to use autografts; a graft preparation station will also be needed. A standard arthroscopic setup with 30- and 70-degree arthroscopes will be necessary for the evaluation and treatment of intra-articular injuries.

Patient Positioning

The patient is placed supine on the operating table and, after administration of an anesthetic, an examination is performed to confirm the suspected ligamentous pathology. A leg holder is placed to allow sufficient access to the medial and lateral aspect of the injured extremity. A well-padded proximal thigh tourniquet is set in place. The operative leg is prepped and draped free in the usual sterile fashion.

Extra-Articular Injury Identification and Treatment

We recommend that open dissection for lateral and/or medial injuries be performed prior to arthroscopic examination, which will allow the identification of injuries and assessment of tissue quality prior to arthroscopic fluid extravasation.

Lateral and Posterolateral Knee

An incision in a hockey-stick shape centered over the posterior to midportion of the iliotibial band is used to expose the lateral/posterolateral knee. The incision is positioned more posteriorly in patients with a planned autogenous patellar tendon graft harvest for a concurrent ACL reconstruction to maintain a minimum of 6 cm between the two incisions. This incision is continued down through the skin and superficial tissues to the superficial layer of the iliotibial band. Posteriorly, the long and short heads of the biceps femoris are identified; palpation approximately 2 to 3 cm distal to the long head will usually allow identification of the common peroneal nerve. A neurolysis is then performed to release the nerve from scar tissue entrapment and safely isolate it from the surgical site (Fig. 103.3).

Avulsions of the biceps tendon are repaired with use of suture anchors with the knee in full extension. Arcuate avulsion fractures may be repaired with No. 5 nonabsorbable sutures passed through the proximal tendon and bony fragment and tied through drill holes in the fibula.[8]

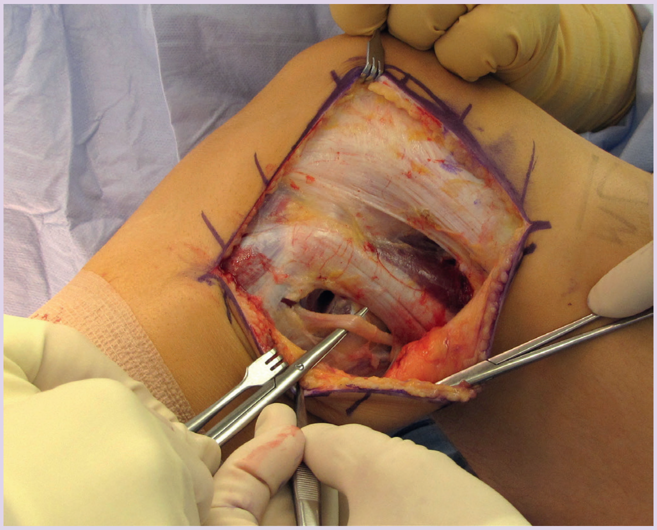

Fig. 103.3 The open surgical approach to the posterolateral aspect of a left knee. The common peroneal nerve is elevated by the Metzenbaum scissors.

Continued

Authors' Preferred Technique

Ligamentous Injuries—cont'd

Blunt dissection between the soleus and the lateral head of the gastrocnemius muscle will expose an interval through which the posteromedial aspect of the fibular head can be palpated. In this region the popliteofibular ligament (PFL) and musculotendinous junction of the PLT are found.

Next, an incision 1 cm proximal to the fibular head is made through the anterior arm of the long head of the biceps and the underlying biceps bursa. The distal aspect and insertion of the FCL can be identified through this incision, and a traction stitch is placed for upcoming identification of the proximal aspect of the ligament. Acute avulsions of the FCL directly from bone without intrasubstance stretch injuries, which usually occur in skeletally immature patients, may be repaired with use of suture anchors; however, the majority of FCL injuries are not amenable to repair and require a ligament reconstruction. A fibular tunnel is drilled with use of a cruciate ligament aiming guide. The desired trajectory for anatomic graft placement is from the FCL attachment site of the lateral aspect of the fibular head to the posteromedial downslope of the fibular styloid. A retractor is placed medially to prevent iatrogenic injury to deep structures, and a guide pin is advanced; the tunnel is overreamed with a 7-mm reamer, and the entry and exit apertures are chamfered with a rasp.

Next, a tibial tunnel is created for passage and fixation of the PFL and PLT reconstruction grafts. The anterior tunnel aperture is located at the flat spot between Gerdy's tubercle and the tibial tubercle; an elevator is used to release the soft tissues from this region. The posterior tunnel aperture is located at the posterolateral aspect of the proximal tibia, slightly distal to the plateau. The previously identified popliteus musculotendinous junction, located 1 cm medial and 1 cm proximal to the fibular head reconstruction tunnel, is the landmark for the posterior aperture of the reconstruction tunnel. A cruciate ligament reconstruction aiming guide is used to create this tunnel; a retractor is placed posteriorly to protect against an erroneously placed guide pin. Accurate placement of the guide pin is confirmed by palpating posteriorly while cross-referencing with a blunt probe placed through the fibular tunnel.

Tension is then applied to the traction stitch in the FCL remnant to identify and evaluate the FCL femoral origin.[74] To allow direct visualization of the FCL and PLT attachment sites on the femur and prepare for potential tunnel drilling, a splitting incision is placed through the superficial layer of the iliotibial band from a point proximal to the lateral epicondyle and extended distally to Gerdy's tubercle. Next, a vertical incision through the lateral capsule allows identification of the femoral insertion of the PLT. Avulsions of the PLT directly from the femur without intrasubstance stretch injury or musculotendinous avulsion may be amenable for a recess procedure performed with the knee in full extension.[75] The creation of femoral tunnels for reconstruction of the FCL and PLT requires a thorough understanding of the anatomy (Fig. 103.4)[74] and is performed according to previously described techniques.[76] With use of a collateral ligament aiming guide, two eyelet-tipped guide pins are aimed anteromedially to the adductor tubercle from the FCL and PLT attachment sites and advanced in a parallel fashion; tunnel orientation is important to avoid collision of the PLC tunnels with ACL reconstruction tunnels. The tunnels are then overreamed to a depth of 20 mm and a diameter of 9 mm (Fig. 103.5).

A split Achilles tendon allograft is prepared for the two limbs of the PLC reconstruction, and the grafts are secured in their femoral tunnels with 7- by 20-mm cannulated interference screws (Fig. 103.6). The FCL graft is passed through the fibular tunnel, but final fixation is delayed until the end of the procedure. Treatment of associated PLC structures is performed when indicated. Popliteomeniscal fascicle and coronary ligament tears are repaired with mattress sutures. Bony (Segond)[18] or soft tissue avulsions of the tibial attachment of the lateral capsular ligament[25] are repaired with suture anchors.[8]

Medial and Posteromedial Knee

The treatment of combined ACL/MCL injuries is well defined and is not specifically discussed here. The focus of this discussion is our preferred treatment of severe grade III medial ligamentous injuries combined with PCL or bicruciate ruptures as well as possible associated lateral injuries. Surgical treatment is delayed until knee swelling decreases; medial tissues may be amenable to repair with augmentation, or a reconstruction may be required. Concurrent, rather than staged, cruciate ligament reconstruction is performed in all patients.

Exposure of the medial knee is performed via an anteromedial incision that extends distally from the region between the medial border of the patella and the medial epicondyle to the region overlying the pes anserine tendons.[70] Next, the gracilis and semitendinosus tendon attachments are identified by incising the anterior border of the sartorial fascia. The semitendinosus tendon is removed with use of a standard tendon harvester and sectioned to create grafts of 16 and 12 cm for reconstruction of the superficial MCL and POL, respectively. Nonabsorbable sutures are used to tubularize each end, and the tendons are sized for 7-mm tunnels.

Within the pes anserine bursa, the superficial MCL distal tibial attachment is identified. Next, the superficial MCL is followed distally, and the tibial attachment is identified at the anteromedial proximal tibia. The superficial MCL has both proximal and distal tibial attachments; the distal attachment is approximately 6 cm distal to the joint line.[65] The tendon of the sartorius muscle is retracted distally and the largest portion of the POL, the central arm, is identified. The tibial attachment site of the POL central arm, with an underlying small bony ridge, can be found near the direct arm of the semimembranosus tendon.

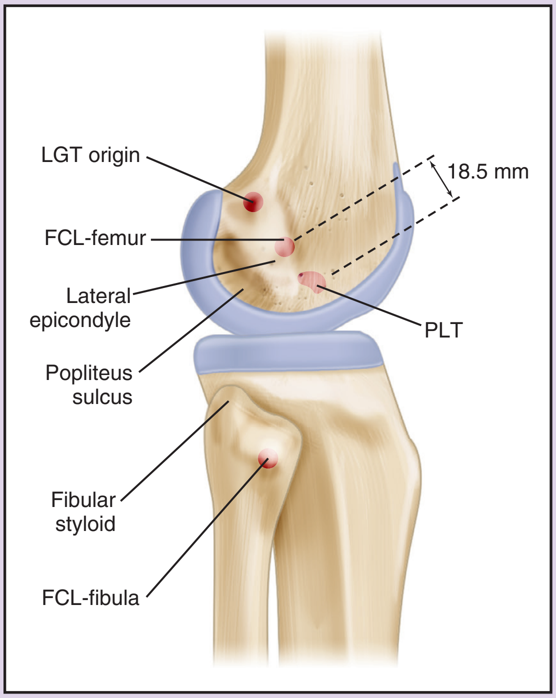

Fig. 103.4 Attachments of key posterolateral knee stabilizing structures and associated bony landmarks. *FCL,* Fibular collateral ligament; *LGT,* lateral gastrocnemius tendon; *PLT,* popliteus tendon.

Authors' Preferred Technique

Ligamentous Injuries—cont'd

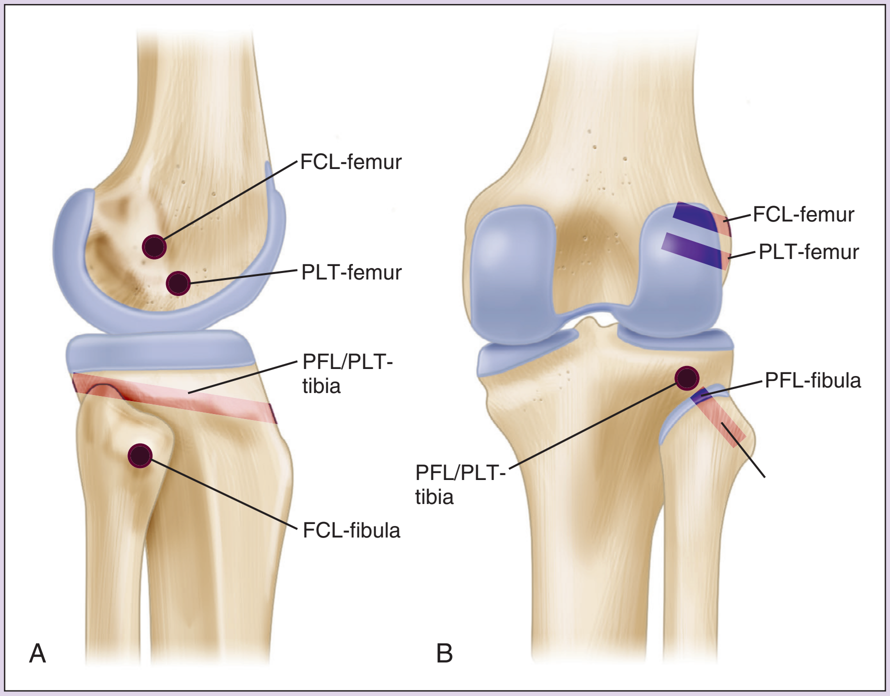

Fig. 103.5 Lateral (A) and posterior (B) aspects of the knee demonstrating the position of bone tunnels for a complete reconstruction of the posterolateral corner of the knee. *FCL,* Fibular collateral ligament; *PFL,* popliteofibular ligament; *PLT,* popliteus tendon.

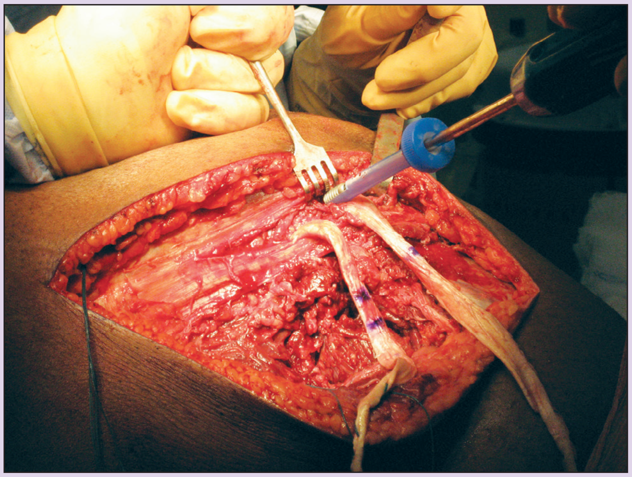

Fig. 103.6 The intraoperative posterolateral aspect of the right knee demonstrating fixation of the posterolateral corner grafts in the femoral tunnels. An iliotibial band–splitting incision is visualized and an interference screw is advanced into the popliteus tendon femoral tunnel.

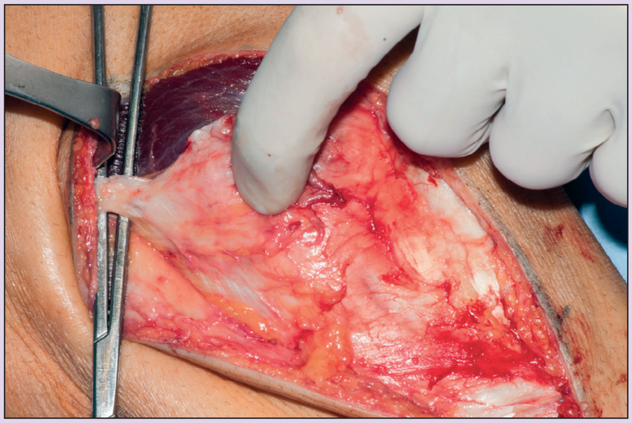

Fig. 103.7 The open surgical approach for the medial aspect of a left knee. The adductor tendon is elevated by the hemostat and a gloved finger is shown palpating the medial epicondyle. The vastus medialis obliquus muscle is also visible.

Identification of the femoral attachments of the medial knee structures is often difficult because the bony and soft tissue landmarks are not as obvious as those on the posterolateral knee. The initial step is to identify the adductor magnus tendon and its distal attachment to the adductor tubercle (Fig. 103.7). The senior author calls the adductor magnus tendon "the lighthouse of the medial aspect of the knee." This step will allow the surgeon to more accurately identify the medial epicondyle, the gastrocnemius tubercle, and the anatomic attachment sites of the superficial MCL and the POL (Fig. 103.8). The femoral attachment of the superficial MCL is approximately 12 mm distal and 8 mm anterior to the adductor tubercle, and the POL femoral attachment is approximately 11 mm posterosuperior to the superficial MCL.[65] After identification of the femoral and tibial attachments of the superficial MCL and POL, guide pins are inserted and overreamed with a 7-mm cannulated drill and advanced to a depth of 30 mm. Allografts or semitendinosus autografts are prepared (16 cm for the superficial MCL and 12 cm for the POL) and are fixed into the femoral tunnels using 7-mm bioabsorbable interference screws. Graft fixation in the tibial tunnels is delayed until the end of the procedure.

Continued

Authors' Preferred Technique

Ligamentous Injuries—cont'd

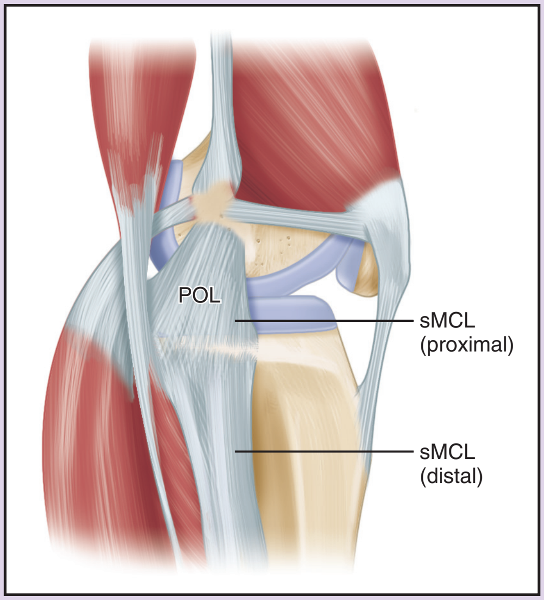

Fig. 103.8 Anatomy of the medial aspect of the left knee. The posterior oblique ligament *(POL)* and superficial medial collateral ligament *(sMCL)* are demonstrated. (From Coobs BR, Wijdicks CA, Armitage BM, et al. An in vitro analysis of an anatomical medial knee reconstruction. *Am J Sports Med*. 2010;38[2]:339–347.)

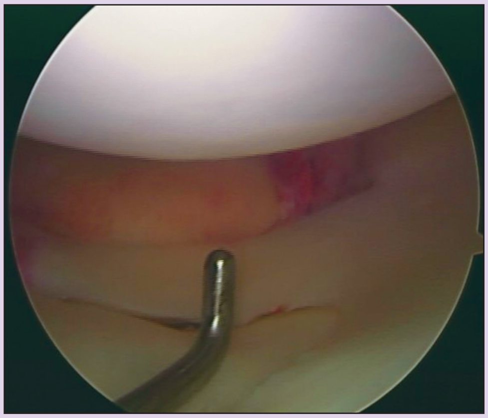

Fig. 103.9 A lateral compartment "drive-through sign" is seen in this arthroscopic photograph in a patient with a posterolateral corner knee injury.

Intra-Articular Injury Identification and Treatment

The arthroscopic portion of the surgery is delayed until after open dissection and injury identification to prevent arthroscopic fluid extravasation. Vertical inferomedial and inferolateral parapatellar portals are created and a standard arthroscopic assessment of the knee is performed. Varus and valgus stress is applied and the lateral and medial compartments, respectively, are observed for gapping (Fig. 103.9). The articular cartilage surfaces are examined for lesions, and the meniscal roots and bodies are similarly assessed; meniscal tears are repaired if possible. Cruciate ligament injuries are addressed as discussed in the following sections.

Posterior Cruciate Ligament

An endoscopic double-bundle PCL reconstruction is preferred in patients with multiligament injuries. An Achilles tendon allograft is used for the ALB reconstruction and a tibialis anterior allograft is used for the PMB reconstruction. The femoral attachments of these two bundles are identified with an arthroscopic coagulator. An additional portal is created through the posteromedial capsule to allow evaluation and debridement of the PCL tibial attachment. Debridement is continued until the popliteus muscle fibers are visualized. With use of a PCL guide, a guide pin is drilled from the anteromedial tibia, approximately 6 cm distal to the joint line, and exits at the PCL tibial attachment approximately 1 cm distal to the joint line at the PCL bundle ridge; pin placement is verified with intraoperative radiographs or fluoroscopy.

Femoral tunnel creation for the PCL reconstruction is performed according to previously described techniques.[59] The ALB is positioned so that the aperture edge is adjacent to the articular cartilage margin at the top of the intercondylar roof and along the anterior aspect of the medial femoral condyle. An 11-mm tunnel is drilled to a depth of 25 mm. Similarly, a 7-mm PMB tunnel is created at the previously described location and reamed to a depth of 25 mm; a minimum 2-mm bone bridge is maintained between the two tunnels.

After creation of the femoral tunnel, the tibial tunnel is reamed. The previously placed tibial guide pin is used to advance a 12-mm headed reamer to create a complete tunnel exiting at the posterior tibia. The posterior tissues are protected against iatrogenic injury via retraction with a large curette placed through the posteromedial portal. To minimize the potential for cyclic graft failure due to friction against the tibial tunnel aperture, a "smoother" is passed through the tibial tunnel and out the anteromedial portal and cycled several times to clean out the posterior tibial aperture.

The grafts are then passed into their respective femoral tunnels endoscopically via the anterolateral arthroscopic portal. A 7-mm titanium screw is used to fix the ALB graft bone plug into its femoral tunnel, and a 7-mm bioabsorbable interference screw is used to fix the PMB soft tissue graft into its femoral tunnel. The ALB and PMB grafts are then pulled distally through the tibial tunnel, and the knee is cycled; tibial graft fixation is delayed until the end of the procedure.

Anterior Cruciate Ligament Reconstruction

A single-bundle anatomic ACL reconstruction is preferred for patients with multiligament injuries. A bone–patellar tendon–bone autograft or allograft is chosen based on the preference of the patient/surgeon and availability. The tibial attachment is identified, and residual tissue is debrided with the shaver. Next, the femoral attachment is similarly identified and debrided to identify the lateral intercondylar ridge. A burr hole is made to mark the midpoint of the ACL femoral attachment between the anteromedial and posterolateral bundles.

The femoral tunnel is created via the anteromedial portal technique. This closed socket femoral tunnel is created with a low-profile reamer prior to tibial tunnel reaming. This sequence allows for identification of the important regional anatomy of this tunnel prior to significant fluid extravasation from the joint. If there was no concurrent medial knee reconstruction incision or if an allograft was used, a 2-cm anteromedial tibial incision is centered approximately 35 mm distal to the joint and 10 mm anterior to the MCL. An ACL reconstruction guide is placed into the joint and centered over the native ACL attachment site; a guide pin is then advanced and the tunnel is reamed. The ACL graft is passed and fixed in the femoral tunnel. Tibial fixation is delayed until the end of the procedure.

Final Graft Fixation

The order of final graft fixation is important. To restore the central pivot of the knee, tibial fixation of the PCL grafts is performed according to previously defined techniques[59] once all associated ligament reconstruction grafts have been fixed

Authors' Preferred Technique

Ligamentous Injuries—cont'd

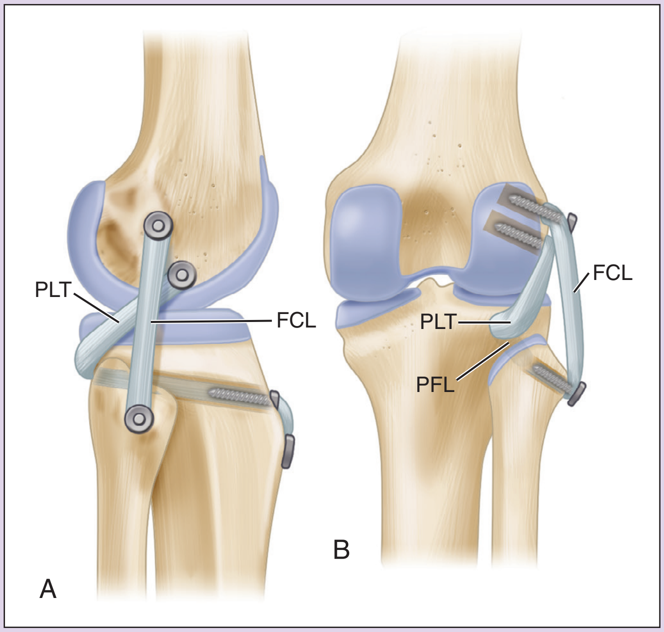

Fig. 103.10 The lateral (A) and posterior (B) views of a posterolateral corner knee reconstruction are demonstrated in this illustration. The fibular collateral ligament *(FCL)*, popliteus tendon *(PLT)*, and popliteofibular ligament *(PFL)* grafts are shown.

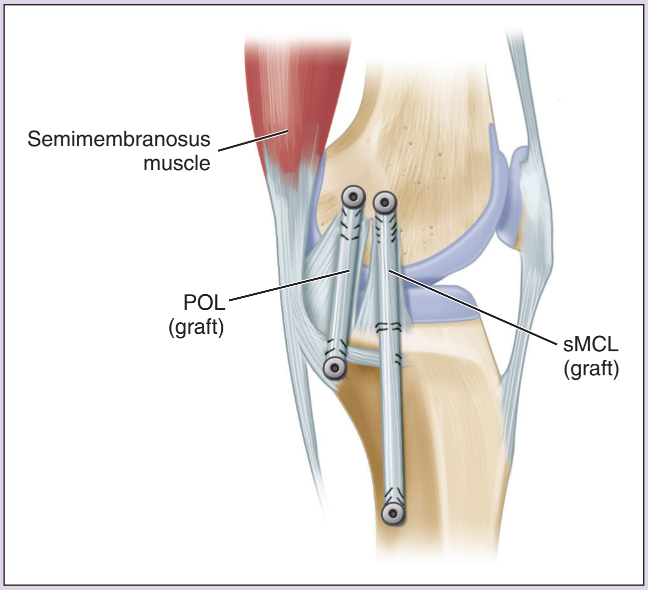

Fig. 103.11 The medial aspect of the left knee with reconstruction grafts fixed in the femoral and tibial tunnels with interference screws. *POL,* Posterior oblique ligament; *sMCL,* superficial medial collateral ligament. (From Coobs BR, Wijdicks CA, Armitage BM, et al. An in vitro analysis of an anatomical medial knee reconstruction. *Am J Sports Med*. 2010;38[2]:339–347.)

in their femoral tunnels. PLC grafts are secured next; care is taken to avoid overreducing the knee in patients with an associated medial knee injury. The FCL graft is passed through the fibula and fixed in its reconstruction tunnel at 20 degrees of knee flexion, in neutral rotation, and with a valgus reduction force on the knee. Next, the PLC grafts are passed anteriorly through the tibial tunnel, slack is removed from the grafts, and fixation of the PLT and PFL grafts is performed at 60 degrees of knee flexion, in neutral rotation, and with traction applied to the grafts (Fig. 103.10).

ACL graft tibial fixation is performed once the PLC grafts have been secured because of biomechanical evidence that fixation of the ACL graft prior to the PLC grafts can result in an external rotation deformity of the knee.[77] An interference screw is used in the tibial tunnel for fixation.

Graft fixation in the tibial tunnels is performed next for patients who underwent reconstruction of medial knee injuries (Fig. 103.11). The superficial MCL graft is passed into the tibial tunnel and tension is held while a varus moment is applied with the knee flexed to 20 degrees and in neutral rotation. At this position, the superficial MCL graft is secured with the interference screw. In a similar fashion, tension is applied to the POL graft via traction in full knee extension. The interference screw is inserted as a varus moment is applied with the knee held in extension and neutral rotation. A suture anchor is then used to recreate the proximal tibial superficial MCL attachment.

POSTOPERATIVE MANAGEMENT

An important consideration in the treatment of multiligament injuries is postoperative rehabilitation and eventual return to play. These topics have received great attention for isolated ligament injuries, but less information is available for patients with multiligament injuries; therefore principles from the former must be applied to the latter. Two general approaches to postoperative care in multiligament injuries are immobilization versus early mobilization. A balance must be achieved between immobilization aimed at preventing the development of instability and early mobilization to minimize scar tissue and resultant range-of-motion deficits. Although a full discussion of various rehabilitation protocols is beyond the scope of this chapter, the basic preferences of several investigators are described here.

Noyes and Barber-Westin described a program of early protected ROM in an attempt to limit arthrofibrosis.[78] They prescribed active assisted motion from 10 to 90 degrees six to eight times daily along with patellar mobilization for the first 4 weeks postoperatively. Between therapy sessions, the patient's knee was immobilized in a split cylinder cast to protect the reconstructed structures. After 4 weeks, the cast was changed to a hinged-brace and gradual increase in motion along with progression of weight bearing was then allowed.

A recent review of rehabilitation after the reconstruction of multiligament injuries recommended complete immobilization with the knee locked in extension for the first 5 weeks after surgery.[79] For the first 6 weeks, the authors allow the patients to bear full weight while standing statically on both legs, but they must use crutches for ambulation and abstain from bearing weight on the operative leg. From postoperative weeks 5 to 10, the brace is unlocked and patients are allowed to perform passive knee flexion; isolated hamstring strengthening is strictly avoided during this period.

Harner et al.[40] also described their preferred surgical treatment and rehabilitation for patients with multiligament injuries. For the first 4 weeks, the knee is held in full extension via a locked knee brace except during passive ROM exercises at up to 90 degrees of flexion. Isolated hamstring contraction is avoided until 6 weeks after surgery. Partial weight bearing is allowed with the brace in full extension for the first 4 to 6 weeks and progresses thereafter.

In a recent systematic review, Mook et al.[36] evaluated the impact of postoperative rehabilitation on multiligament injury outcomes. Their findings indicate that early mobilization in acutely treated patients is associated with less posterior instability, varus/valgus laxity, and ROM deficits as compared with immobilization after surgery. The apparent influence of early mobilization underscores the importance of a strong anatomic repair and/or reconstruction of posterolateral structures.

Wilkins[80] provided a detailed description of rehabilitation principles based on work by LaPrade and colleagues and describes early-phase rehabilitation (0 to 12 weeks) and late-phase rehabilitation (4 to 12 months). Similar to other authors, Wilkins recommends non–weight bearing for the first 6 weeks postoperatively, with occasional toe-touch weight bearing during showering and dressing activities; weight bearing is thereafter increased according to the protocol. For the first 2 weeks the brace is locked in full extension with the exception of therapy sessions, where the patient is allowed knee motion within the "safe zone" as defined intraoperatively. ROM is then increased to 90 degrees until 4 weeks postoperatively and thereafter gradually progressed to the goal of greater than or equal to 125 degrees. Because of posterior tibial translation associated with increasing flexion, isolated hamstring exercises are avoided for the first 4 months. Late-phase rehabilitation focuses on high-level balance training, sport-specific drills, and plyometric exercises and is initiated at approximately 4 months after surgery, when the patient has regained full active ROM, has resumed a normal gait, and has no signs of swelling.

RESULTS

No level I studies are available on the topic of treatment and associated outcomes of surgical reconstruction of multiligament injuries. Most available studies are level III or IV, and unfortunately evaluation of the results of some studies is limited by the heterogeneous patient mix. Often isolated injuries were included in the analysis with multiligament injuries, and high-energy injuries in polytrauma patients may be reported along with low-energy injuries associated with sports. A summary of selected literature is provided in Table 103.1.

COMPLICATIONS

Complications can be discussed within the context of the initial injury or those associated with treatment. Vascular injuries are not infrequent in the setting of multiligament injuries, and although the reported incidence varies, a commonly referenced number is 32%.[81] Urgent vascular surgery consultation is required for suspected large vessel injury because prolonged ischemia may necessitate limb amputation. Peroneal nerve injuries are also common with multiligament injuries, especially when they are combined with PLC injuries, and are estimated to occur at a rate of 25% to 35%.[7] A poor prognosis is associated with complete lesions, whereas most persons with incomplete nerve palsies can often be expected to recover. Treatment options include physical therapy with an ankle-foot orthosis, neurolysis, primary nerve repair, nerve grafting, and tendon transfers.[82] With high-energy injuries, infection and skin compromise may also occur; inevitably definitive surgery will be delayed while skin concerns are addressed.

Complications may also be associated with treatment, whether nonoperative or operative. In both cases, persistent pain and instability may occur. If the initial treatment was nonoperative, the treatment is deemed a failure, and if the patient is a candidate, a chronic reconstruction may be required. If the initial treatment was operative, the patient may require revision reconstruction of all failed components.

Infection and bleeding can also occur with operative treatment. Incision and debridement with antibiotics and staged reconstruction may be necessary for infected cases. Bleeding may be the expected result of a difficult exposure, but attempts at adequate hemostasis must be obtained prior to closure. Bleeding may also occur because of iatrogenic injury to large vessels such as the popliteal artery that necessitates immediate intervention by a vascular surgeon. Iatrogenic injury to the common peroneal nerve may occur during neurolysis or as a failure of an adequate neurolysis prior to ligament reconstruction. However, adequately exposed and carefully handled nerves may still be associated with foot drop if the initial injury was severe. As with any surgery, deep venous thrombosis may occur; adequate prophylaxis is essential, especially in older and obese patients.

General causes of ligament reconstruction failure may include lack of incorporation of allografts, unrecognized or untreated ligament lesions, or technical concerns due to the repair technique or reconstruction tunnel placement. A persistent postoperative effusion may occur in some patients and will necessitate delaying progression of the rehabilitation protocol.

FUTURE CONSIDERATIONS

Because of the rarity of multiligament injuries and the variation of severity and presence of concomitant injuries, studies with a high level of evidence are limited. Additional clinical studies are necessary to make definitive recommendations on operative timing, surgical technique, and rehabilitation. Level I studies with randomization of surgical treatment may seem impractical because of the complexity of these injuries and the treatment preferences of particular surgeons. However, prospective multicenter studies with treatment outcomes assessed by a standardized method with validated subjective evaluations and unbiased objective measurements of stability and function would significantly benefit the multiligament injury evidence base.

In the past one to two decades, several ligament reconstruction techniques have been developed and tested in persons with isolated ligament injuries. However, few studies address the biomechanics of multiligament injuries and subsequent ligament

TABLE 103.1 Summary of Selected Recent Literature on Multiligament Knee Injuries

Study	Year	No. of Knees	Male (%)	Years of Follow-Up, Average (Range)	Timing: No. of Patients (Time to Surgery)	Average Age (Years)	Injury Type (Schenck Classification)	Knee Function (Average Scores)	Physical Examination (Stability)
Levy	2010	28	Not given	Repair: 2.8 (2–4.1) Reconstruction: 2.3 (2–3.4)	Repair: 10 (5–33 days) Reconstruction: 18 (17–731 days)	*	KD I: 12 KD III: 10 KD IV: 44 KD V: 1	4/10 repairs failed 1/18 reconstructions failed IKDC: 79 (repair), 77 (reconstruction) Lysholm: 85 (repair), 88 (reconstruction)	IKDC objective at final follow-up: 0 repairs and 2 reconstructions had 1+ laxity to varus stress at 30 degrees All patients were stable with the dial test at 30 and 90 degrees
Engebretsen	2009	85	53	5.3 (2–9.9)	A: 50 (<2 week) C: 35 Average: 14 months	35.2	KD II–III: 88% KD IV: 12%	Lysholm: 83 IKDC 2000: 64	Lachman: 57% normal, 33% 1+ Pivot shift: 90% normal, 4% 1+ Posterior drawer: 26% normal, 57% 1+ Dial test: 86% normal, 11% 1+ Valgus: 60% normal, 30% 1+ Varus: 67% normal, 25% 1+
Strobel	2006	17	76	(2–5.5)	C: 17 (5–312 months)	30.7	KD III: 17	IKDC: 71.8	Posterior drawer: 88% grade I or II, 12% grade III Lachman: 94% grade I or II, 6% grade III Posterolateral instability: 88% normal or nearly normal; 12% with residual instability
Stannard	2005	57	63	2.8 (2–4.9)	35 patients qualified for repair and were treated within 3 wk; 22 patients were chronic or had nonrepairable acute injuries	33	44 multiligament 13 isolated	Mean Lysholm, IKDC subjective repair success: 88.2, 59.8 Failed repair, s/p revision: 86.8, 63.6 Reconstruction success: 89.6, 56.1 Failed reconstruction, s/p revision: 92, 64.4	13/35 repairs failed; 12 underwent successful revision reconstruction 2/22 reconstructions failed IKDC objective: PLC repair success: 4 A, 14 B, 2 C, 1 D PLC repair failure: 3 A, 4 B, 4 C, 0 D PLC reconstruction success: 7 A, 9 B, 2 C, 2 D PLC reconstruction failure: 0 A, 1 B, 1 C, 0 D
Harner	2004	31	*	3.7 (2–6)	A: 19 (within 3 week) C: 12 (5 week–22 months)	28.4	Acute: KD I: 3 KD II+: 16 Chronic: KD II: 7 KD III-L: 2 KD III-M: 3	Lysholm A: 91; C: 80 KOS ADL A: 91; C 84 Sports Activities Scale: A: 89; C: 69 Meyers: 10 excellent, 13 good, 5 fair, 3 poor	Lachman: 48% normal, 52% 1+ Posterior drawer: 71% 1+, 29% 2+ Varus stress: 30 degrees: 29% 1+, 6% 2+ Valgus stress 30°: 16% 1+, 13% 2+
Fanelli	2002	35	74	(2–10)	A: 19 (<8 week) C: 16 (3–26 months)	*	KD II: 1 KD III-L: 19 KD III-M: 9 KD IV: 6	Lysholm: 91.2 HSS: 86.8	Lachman and pivot shift: 94% normal Posterior drawer: 46% normal, 54% 1+ Posterolateral stability: normal in 24%, "tighter than normal" in 76%

*Not given.
A, Acute; *C,* chronic; *HSS,* Hospital for Special Surgery; *IKDC,* International Knee Documentation Committee; *KD,* knee dislocation; *KOS ADLA,* Knee Outcome Survey Activities of Daily Living Scale; *PLC,* posterolateral corner; *s/p,* status postoperative.

reconstruction. Additionally, laboratory and clinical studies may evaluate the role of biologic treatments such as the use of stem cells and applied growth factors including platelet-rich plasma therapy in the setting of multiligament reconstruction.

For a complete list of references, go to ExpertConsult.com.

SELECTED READINGS

Citation:
Mook WR, Miller MD, Diduch DR, et al. Multiple-ligament knee injuries: a systematic review of the timing of operative intervention and postoperative rehabilitation. *J Bone Joint Surg Am*. 2009;91A(12):2946–2957.

Level of Evidence:
III

Summary:
Mook and colleagues performed a systematic review of the literature on multiple-ligament knee injuries to assess the impact of surgical timing and postoperative rehabilitation. The study included 24 retrospective studies and revealed several findings regarding the impact of these two factors on subjective and objective outcomes, which are discussed, with the associated limitations.

Citation:
Engebretsen L, Risberg MA, Robertson B, et al. Outcome after knee dislocations: a 2-9 years follow-up of 85 consecutive patients. *Knee Surg Sports Traumatol Arthrosc*. 2009;17(9):1013–1026.

Level of Evidence:
IV

Summary:
Engebretsen and colleagues report outcomes and surgical technique for 85 patients after knee dislocation. At final follow-up, the median Lysholm score was 83, knee function was found to be lower in patients who sustained high-energy compared with low-energy trauma, and 87% of injured knees had Kellgren and Lawrence grade 2 or higher compared with 35% of the uninjured knees.

Citation:
Harner CD, Waltrip RL, Bennett CH, et al. Surgical management of knee dislocations. *J Bone Joint Surg Am*. 2004;86A(2):262–273.

Level of Evidence:
III

Summary:
The authors describe their surgical technique and associated outcomes for 33 patients with multiligament knee injuries. Importantly, they found that acutely treated patients had higher subjective scores and improved objective knee stability.

Citation:
Fanelli GC, Edson CJ. Arthroscopically assisted combined anterior and posterior cruciate ligament reconstruction in the multiple ligament injured knee: 2- to 10-year follow-up. *Arthroscopy*. 2002;18(7):703–714.

Level of Evidence:
III

Summary:
Surgical technique and associated outcomes for 35 patients with multiligament injuries are reported. Nineteen patients were treated within 8 weeks and 16 were treated chronically; no significant subjective or objective differences were found between the two groups.

Citation:
Stannard JP, Brown SL, Farris RC, et al. The posterolateral corner of the knee: repair versus reconstruction. *Am J Sports Med*. 2005;33(6):881–888.

Level of Evidence:
II

Summary:
Stannard and colleagues performed a cohort study comparing posterolateral corner (PLC) repair versus reconstruction in combination with treatment of multiligament knee injuries. The failure rate for repairs of the PLC was significantly higher than for reconstructions.

REFERENCES

1. Engebretsen L, Risberg MA, Robertson B, et al. Outcome after knee dislocations: a 2-9 years follow-up of 85 consecutive patients. *Knee Surg Sports Traumatol Arthrosc*. 2009;17(9):1013–1026.
2. Moatshe G, Dornan GJ, Loken S, et al. Demographics and injuries associated with knee dislocation. A prospective review of 303 patients. *Orthop J Sports Med*. 2017;5:5.
3. Moatshe G, Dornan GJ, Ludvigsen TC, et al. High prevalence of knee osteoarthritis at a minimum 10-year follow-up after knee dislocation surgery. *Knee Surg Sports Traumatol Arthrosc*. 2017.
4. Shelbourne KD, Klootwyk TE. Low-velocity knee dislocation with sports injuries. Treatment principles. *Clin Sports Med*. 2000;19(3):443–456.
5. Mills WJ, Barei DP, McNair P. The value of the ankle-brachial index for diagnosing arterial injury after knee dislocation: a prospective study. *J Trauma*. 2004;56(6):1261–1265.
6. Fanelli GC, Stannard JP, Stuart MJ, et al. Management of complex knee ligament injuries. *J Bone Joint Surg Am*. 2010;92(12):2235–2246.
7. Johnson ME, Foster L, DeLee JC. Neurologic and vascular injuries associated with knee ligament injuries. *Am J Sports Med*. 2008;36(12):2448–2462.
8. Geeslin AG, LaPrade RF. Outcomes of treatment of acute grade-III isolated and combined posterolateral knee injuries: a prospective case series and surgical technique. *J Bone Joint Surg Am*. 2011;93(18):1672–1683.
9. Geeslin AG, LaPrade RF. Location of bone bruises and other osseous injuries associated with acute grade III isolated and combined posterolateral knee injuries. *Am J Sports Med*. 2010;38(12):2502–2508.
10. LaPrade RF, Terry GC. Injuries to the posterolateral aspect of the knee. Association of anatomic injury patterns with clinical instability. *Am J Sports Med*. 1997;25(4):433–438.
11. LaPrade RF, Heikes C, Bakker AJ, et al. The reproducibility and repeatability of varus stress radiographs in the assessment of isolated fibular collateral ligament and grade-III posterolateral knee injuries. An in vitro biomechanical study. *J Bone Joint Surgery Am*. 2008;90(10):2069–2076.
12. LaPrade RF, Ly TV, Griffith C. The external rotation recurvatum test revisited: reevaluation of the sagittal plane tibiofemoral relationship. *Am J Sports Med*. 2008;36(4):709–712.
13. Bae JH, Choi IC, Suh SW, et al. Evaluation of the reliability of the dial test for posterolateral rotatory instability: a cadaveric study using an isotonic rotation machine. *Arthroscopy*. 2008;24(5):593–598.
14. Strauss EJ, Ishak C, Inzerillo C, et al. Effect of tibial positioning on the diagnosis of posterolateral rotatory instability in the posterior cruciate ligament-deficient knee. *Br J Sports Med*. 2007;41(8):481–485.
15. LaPrade RF, Bernhardson AS, Griffith CJ, et al. Correlation of valgus stress radiographs with medial knee ligament injuries: an in vitro biomechanical study. *Am J Sports Med*. 2010;38(2):330–338.
16. LaPrade RF, Wijdicks CA. The management of injuries to the medial side of the knee. *J Orthop Sports Phys Ther*. 2012;42(3): 221–233.
17. Rosenberg TD, Paulos LE, Parker RD, et al. The forty-five-degree posteroanterior flexion weight-bearing radiograph of the knee. *J Bone Joint Surg Am*. 1988;70(10):1479–1483.
18. Segond P: Recherches cliniques et experimentales sur les epanchements sanguins du genou par entorse, Paris, 1879, *Progres Med VII*, pp 1–84.
19. Shindell R, Walsh WM, Connolly JF. Avulsion fracture of the fibula: the "arcuate sign" of posterolateral knee instability. *Nebr Med J*. 1984;69(11):369–371.
20. LaPrade RF, Johansen S, Agel J, et al. Outcomes of an anatomic posterolateral knee reconstruction. *J Bone Joint Surgery Am*. 2010;92(1):16–22.
21. Noyes FR, Barber-Westin SD. Posterior cruciate ligament revision reconstruction, part 1: causes of surgical failure in 52 consecutive operations. *Am J Sports Med*. 2005;33(5):646–654.
22. Jackman T, LaPrade RF, Pontinen T, et al. Intraobserver and interobserver reliability of the kneeling technique of stress radiography for the evaluation of posterior knee laxity. *Am J Sports Med*. 2008;36(8):1571–1576.
23. Schulz MS, Russe K, Lampakis G, et al. Reliability of stress radiography for evaluation of posterior knee laxity. *Am J Sports Med*. 2005;33(4):502–506.
24. Hewett TE, Noyes FR, Lee MD. Diagnosis of complete and partial posterior cruciate ligament ruptures. Stress radiography compared with KT-1000 arthrometer and posterior drawer testing. *Am J Sports Med*. 1997;25(5):648–655.
25. LaPrade RF, Gilbert TJ, Bollom TS, et al. The magnetic resonance imaging appearance of individual structures of the posterolateral knee. A prospective study of normal knees and knees with surgically verified grade III injuries. *Am J Sports Med*. 2000;28(2):191–199.
26. Graf BK, Cook DA, De Smet AA, et al. "Bone bruises" on magnetic resonance imaging evaluation of anterior cruciate ligament injuries. *Am J Sports Med*. 1993;21(2):220–223.
27. Miller MD, Osborne JR, Gordon WT, et al. The natural history of bone bruises. A prospective study of magnetic resonance imaging-detected trabecular microfractures in patients with isolated medial collateral ligament injuries. *Am J Sports Med*. 1998;26(1):15–19.
28. Miller MD, Osborne JR, Gordon WT, et al. The natural history of bone bruises: A prospective study of magnetic resonance imaging-detected trabecular microfractures in patients with isolated medial collateral ligament injuries. *Am J Sports Med*. 1998;26(1):15–19.
29. Lunden JB, Bzdusek PJ, Monson JK, et al. Current concepts in the recognition and treatment of posterolateral corner injuries of the knee. *J Orthop Sports Phys Ther*. 2010;40(8):502–516.
30. Fanelli GC, Orcutt DR, Edson CJ. The multiple-ligament injured knee: evaluation, treatment, and results. *Arthroscopy*. 2005;21(4): 471–486.
31. Hughston JC, Andrews JR, Cross MJ, et al. Classification of knee ligament instabilities. Part I. The medial compartment and cruciate ligaments. *J Bone Joint Surg Am*. 1976;58(2):159–172.
32. Hughston JC, Andrews JR, Cross MJ, et al. Classification of knee ligament instabilities. Part II. The lateral compartment. *J Bone Joint Surg Am*. 1976;58(2):173–179.
33. American Medical Association. Standard nomenclature of athletic injuries. In: *Committee on the Medical Aspects of Sports*. Chicago: American Medical Association; 1968.
34. Schenck RJ. Classification of knee dislocations. *Oper Tech Sports Med*. 2003;11:193–198.
35. Levy BA, Fanelli GC, Whelan DB, et al. Controversies in the treatment of knee dislocations and multiligament reconstruction. *J Am Acad Orthop Surg*. 2009;17(4):197–206.
36. Mook WR, Miller MD, Diduch DR, et al. Multiple-ligament knee injuries: a systematic review of the timing of operative intervention and postoperative rehabilitation. *J Bone Joint Surg Am*. 2009;91(12):2946–2957.

37. Peskun CJ, Whelan DB. Outcomes of operative and nonoperative treatment of multiligament knee injuries: an evidence-based review. *Sports Med Arthrosc*. 2011;19(2):167–173.
38. Kennedy JC. Complete dislocation of the knee joint. *J Bone Joint Surg Am*. 1963;45:889–904.
39. Taylor AR, Arden GP, Rainey HA. Traumatic dislocation of the knee. A report of forty-three cases with special reference to conservative treatment. *J Bone Joint Surg Br*. 1972;54(1):96–102.
40. Harner CD, Waltrip RL, Bennett CH, et al. Surgical management of knee dislocations. *J Bone Joint Surg Am*. 2004;86-A(2): 262–273.
41. Fanelli GC, Edson CJ. Arthroscopically assisted combined anterior and posterior cruciate ligament reconstruction in the multiple ligament injured knee: 2- to 10-year follow-up. *Arthroscopy*. 2002;18(7):703–714.
42. Arthur A, LaPrade RF, Agel J. Proximal tibial opening wedge osteotomy as the initial treatment for chronic posterolateral corner deficiency in the varus knee: a prospective clinical study. *Am J Sports Med*. 2007;35(11):1844–1850.
43. Noyes FR, Barber-Westin SD, Albright JC. An analysis of the causes of failure in 57 consecutive posterolateral operative procedures. *Am J Sports Med*. 2006;34(9):1419–1430.
44. Veltri DM, Deng XH, Torzilli PA, et al. The role of the cruciate and posterolateral ligaments in stability of the knee. A biomechanical study. *Am J Sports Med*. 1995;23(4):436–443.
45. Levy BA, Dajani KA, Morgan JA, et al. Repair versus reconstruction of the fibular collateral ligament and posterolateral corner in the multiligament-injured knee. *Am J Sports Med*. 2010;38(4):804–809.
46. Stannard JP, Brown SL, Farris RC, et al. The posterolateral corner of the knee: repair versus reconstruction. *Am J Sports Med*. 2005;33(6):881–888.
47. LaPrade RF, Resig S, Wentorf F, et al. The effects of grade III posterolateral knee complex injuries on anterior cruciate ligament graft force. A biomechanical analysis. *Am J Sports Med*. 1999;27(4):469–475.
48. Harner CD, Vogrin TM, Hoher J, et al. Biomechanical analysis of a posterior cruciate ligament reconstruction. Deficiency of the posterolateral structures as a cause of graft failure. *Am J Sports Med*. 2000;28(1):32–39.
49. LaPrade RF, Muench C, Wentorf F, et al. The effect of injury to the posterolateral structures of the knee on force in a posterior cruciate ligament graft: a biomechanical study. *Am J Sports Med*. 2002;30(2):233–238.
50. Geeslin AG, Moulton SG, LaPrade RF. A systematic review of the outcomes of posterolateral corner knee injuries, Part 1: surgical treatment of acute injuries. *Am J Sports Med*. 2016;44(5): 1336–1342.
51. McConkey MO, Bonasia DE, Amendola A. Pediatric anterior cruciate ligament reconstruction. *Curr Rev Musculoskelet Med*. 2011;4(2):37–44.
52. Bovid KM, Salata MJ, Vander Have KL, et al. Arthroscopic posterior cruciate ligament reconstruction in a skeletally immature patient: a new technique with case report. *Arthroscopy*. 2010;26(4):563–570.
53. Williams BT, James EW, LaPrade RF. A physeal-sparing fibular collateral ligament and proximal tibiofibular joint reconstruction in a skeletally immature athlete. *Knee Surg Sports Traumatol Arthrosc*. 2016;24:661–665.
54. Azar FM, Brandt JC, Miller RH, et al. Ultra-low-velocity knee dislocations. *Am J Sports Med*. 2011;39(10):2170–2174.
55. Werner BC, Gwathmey FW, Higgins ST, et al. Ultra-low velocity knee dislocations: patient characteristics, complications, and outcomes. *Am J Sports Med*. 2014;42(2):358–363.
56. Shelbourne KD, Porter DA, Clingman JA, et al. Low-velocity knee dislocation. *Orthop Rev*. 1991;20(11):995–1004.
57. Strobel MJ, Schulz MS, Petersen WJ, et al. Combined anterior cruciate ligament, posterior cruciate ligament, and posterolateral corner reconstruction with autogenous hamstring grafts in chronic instabilities. *Arthroscopy*. 2006;22(2):182–192.
58. Race A, Amis AA. PCL reconstruction. In vitro biomechanical comparison of "isometric" versus single and double-bundled "anatomic" grafts. *J Bone Joint Surg Br*. 1998;80(1):173–179.
59. Spiridonov SI, Slinkard NJ, LaPrade RF. Isolated and combined grade-III posterior cruciate ligament tears treated with double-bundle reconstruction with use of endoscopically placed femoral tunnels and grafts: operative technique and clinical outcomes. *J Bone Joint Surg Am*. 2011;93(19):1773–1780.
60. Chhabra A, Cha PS, Rihn JA, et al. Surgical management of knee dislocations. Surgical technique. *J Bone Joint Surg Am*. 2005;87(suppl 1 Pt 1):1–21.
61. May JH, Gillette BP, Morgan JA, et al. Transtibial versus inlay posterior cruciate ligament reconstruction: an evidence-based systematic review. *J Knee Surg*. 2010;23(2):73–79.
62. Noyes FR, Barber-Westin SD. Two-strand posterior cruciate ligament reconstruction with a quadriceps tendon-patellar bone autograft: technical considerations and clinical results. *Instr Course Lect*. 2006;55:509–528.
63. Stannard JP, Riley RS, Sheils TM, et al. Anatomic reconstruction of the posterior cruciate ligament after multiligament knee injuries. A combination of the tibial-inlay and two-femoral-tunnel techniques. *Am J Sports Med*. 2003;31(2):196–202.
64. Cooper DE, Stewart D. Posterior cruciate ligament reconstruction using single-bundle patella tendon graft with tibial inlay fixation: 2- to 10-year follow-up. *Am J Sports Med*. 2004;32(2):346–360.
65. LaPrade RF, Engebretsen AH, Ly TV, et al. The anatomy of the medial part of the knee. *J Bone Joint Surgery Am*. 2007;89(9): 2000–2010.
66. Kovachevich R, Shah JP, Arens AM, et al. Operative management of the medial collateral ligament in the multi-ligament injured knee: an evidence-based systematic review. *Knee Surg Sports Traumatol Arthrosc*. 2009;17(7):823–829.
67. Fanelli GC, Giannotti BF, Edson CJ. Arthroscopically assisted combined anterior and posterior cruciate ligament reconstruction. *Arthroscopy*. 1996;12(1):5–14.
68. Lind M, Jakobsen BW, Lund B, et al. Anatomical reconstruction of the medial collateral ligament and posteromedial corner of the knee in patients with chronic medial collateral ligament instability. *Am J Sports Med*. 2009;37(6):1116–1122.
69. Coobs BR, Wijdicks CA, Armitage BM, et al. An in vitro analysis of an anatomical medial knee reconstruction. *Am J Sports Med*. 2010;38(2):339–347.
70. LaPrade RF, Wijdicks CA. Surgical technique: development of an anatomic medial knee reconstruction. *Clin Orthop Relat Res*. 2012;470(3):806–814.
71. Baker CL Jr, Norwood LA, Hughston JC. Acute posterolateral rotatory instability of the knee. *J Bone Joint Surgery Am*. 1983;65(5):614–618.
72. Baker CL Jr, Norwood LA, Hughston JC. Acute combined posterior cruciate and posterolateral instability of the knee. *Am J Sports Med*. 1984;12(3):204–208.

73. DeLee JC, Riley MB, Rockwood CA Jr. Acute posterolateral rotatory instability of the knee. *Am J Sports Med*. 1983;11(4): 199–207.
74. LaPrade RF, Ly TV, Wentorf FA, et al. The posterolateral attachments of the knee: a qualitative and quantitative morphologic analysis of the fibular collateral ligament, popliteus tendon, popliteofibular ligament, and lateral gastrocnemius tendon. *Am J Sports Med*. 2003;31(6):854–860.
75. LaPrade RF, Hamilton CD, Engebretsen L. Treatment of acute and chronic combined anterior cruciate ligament and posterolateral knee ligament injuries. *Sports Med Arthrosc*. 1997;5(2):91–99.
76. LaPrade RF, Johansen S, Wentorf FA, et al. An analysis of an anatomical posterolateral knee reconstruction: an in vitro biomechanical study and development of a surgical technique. *Am J Sports Med*. 2004;32(6):1405–1414.
77. Wentorf FA, LaPrade RF, Lewis JL, et al. The influence of the integrity of posterolateral structures on tibiofemoral orientation when an anterior cruciate ligament graft is tensioned. *Am J Sports Med*. 2002;30(6):796–799.
78. Noyes FR, Barber-Westin SD. Reconstruction of the anterior and posterior cruciate ligaments after knee dislocation. Use of early protected postoperative motion to decrease arthrofibrosis. *Am J Sports Med*. 1997;25(6):769–778.
79. Edson CJ, Fanelli GC, Beck JD. Rehabilitation after multiple-ligament reconstruction of the knee. *Sports Med Arthrosc*. 2011;19(2):162–166.
80. Wilkins SA. Rehabilitation of the multiple ligament injured knee. *Tech Knee Surg*. 2011;10(1):2–10.
81. Green NE, Allen BL. Vascular injuries associated with dislocation of the knee. *J Bone Joint Surg Am*. 1977;59(2):236–239.
82. Giuseffi SA, Bishop AT, Shin AY, et al. Surgical treatment of peroneal nerve palsy after knee dislocation. *Knee Surg Sports Traumatol Arthrosc*. 2010;18(11):1583–1586.